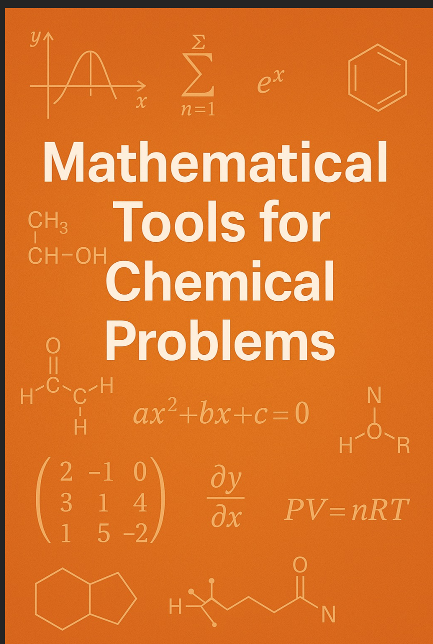

CHEM 3PC3 F2026

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McMaster University

GenAI for Quantum Science

Chem.AI Lab

AI for Materials

- How much can ML models learn?
- Can we improve physical models?
- Do pre-trained ML models work?

Housekeeping

Class Schedule	**Wednesday 9:30AM - 10:20AM** **Friday 8:30AM - 10:20AM**
Location	**MDCL 1116**
Tutorial Schedule	Wednesday 11:30 AM – 12:20 PM
Location	BSB B142
Office Hours	TBD
Location	TBD (probably ABB 266)

*dates could change during the term

Course Overview and Assessment*

Week	Dates	Topic	Book	Chapter
1	Sept. 7-11	Univariate Calculus and Series:	PC	1
2	Sept. 14-18	Thermodynamics, heat capacity, and integration	MPC	1,2
3	Sept. 21-25	Multivariate Calculus: Fundamental Equations of Thermodynamics	PC	2,3
4	Sept. 28- Oct. 2		AMPC MPC	4,7 12
5	Oct. 5 - Oct. 9 and Oct 21	Linear Algebra: Simple and complex equilibrium	LAWA	1
7	**Oct. 12-18**	**Fall break. No classes this week!!**		**None**
	Oct 23	**Midterm exam (In Class)**		
8	Oct. 26-30	Linear Regression	LAWA	2,3
9	Nov. 2 - 6	Linear Regression	PC	9
10	Nov. 9 - 13	Non Linear Equations	MPC	16
11	Nov. 16 - 20	Non Linear Equations	MPC	19
12	Nov. 23 - 27	Eigenvalues and eigenvectors	LAWA	7
13	Nov. 30 - Dec. 4	Eigenvalues and eigenvectors	NMSE	7
14	Dec. 9	Molecular Orbitals	Class notes	None
	TBD	Final exam		

Week structure

Week	Topic
Wednesday	Theory/Math Lecture
Friday	Theory/Math + Programming Lecture

https://chemai-lab.github.io/Math4Chem

*link will be posted on AV2L :/

Tutorials

Tutorial Schedule	Wednesday 11:30 AM – 12:20 PM
Location	BSB B142

Alexandre de Camargo

Goals:

1. Answer **YOUR** questions.
2. HELP you understand the content.
3. Codding and math exercises.
4. **Weekly quizzes**

Until Sept 24th tutorials are online

Zoom link: https://mcmaster.zoom.us/j/96863889054
Meeting ID: 968 6388 9054
Passcode: 721109

Evaluation

Assessment Method	Weight
Four Avenue Quizzes (dates TBD)	20%
2 Python assignments (dates TBD)	20%
Mid-Term Exam	25%
Final Exam	35%
Total	100%

If your final exam's mark is higher, I will give you that mark!

With great power comes great responsibility

Python Assignments

I believe that in today's job market one of the most underrated skills is the ability to work with different people.

- You will work in groups of 2 (3).
- Submission will be a *.ipynb* file deployable in GoogleColab.
  - The teams will be randomly assigned this week when enrollment is final.
- STUDENT1_SURNAME1_STUDENT1_ID_ STUDENT2_SURNAME2_ STUDENT1_ID _QUIZ_NUMBER.ipynb (e.g., SMITH_001234567_JUAN_ 001234098_QUIZ_2.ipynb).

Books and additional resources.

Books:

- Applied mathematics for Physical Chemistry,
  James R. Barrante.
- Mathematics for Physical Chemistry,
  Donald A. McQuarrie.
- Physical Chemistry,
  Ira N. Levine.
- Linear Algebra with applications,
  Gareth Williams.
- The Matrix Cookbook,
  http://matrixcookbook.com

Other resources

- Use the internet.
  - YouTube.
  - https://chem.libretexts.org/
  - https://stackoverflow.com/
- Notes from prev. years will be provided.

Course and Learning Objectives

At the end of the course, the student will have elementary algebra, calculus, and programming skills tailored to model chemical systems.

QUIZ 0

Try to answer as much questions as possible.

Do not write your name.

This quiz does not count towards your mark.

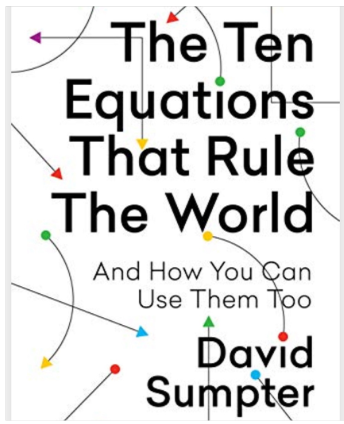
Academic Integrity in the era of Generative AI

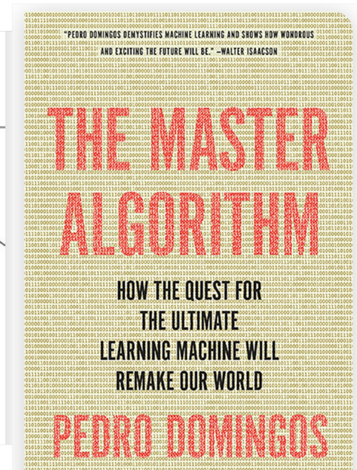
As a researcher in the field of machine learning and AI, I recognize the significant potential of these modern tools to enhance the learning process. I support their responsible use, with the understanding that the author who submits the work is ultimately accountable for its content and any potential repercussions.
In this course, where the primary objective is to develop students' mathematical and coding skills, it is essential to ensure that all work reflects genuine analysis and understanding.

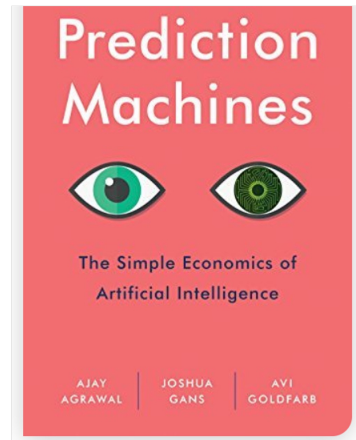
The instructor and TA will be vigilant in identifying any instances of **"copy-paste" answers**, particularly those that appear to be generated by AI without sufficient student comprehension. If the instructor or TA suspects that an answer has been directly copied from a chatbot, the student will be required to explain their reasoning. **Responses such as "this is what ChatGPT told me" will not be tolerated. Based on the explanation provided, appropriate consequences may be determined.**

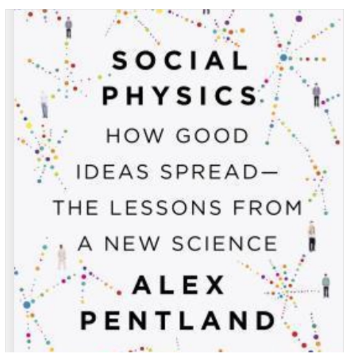
Use these amazing tools but do not take them for granted

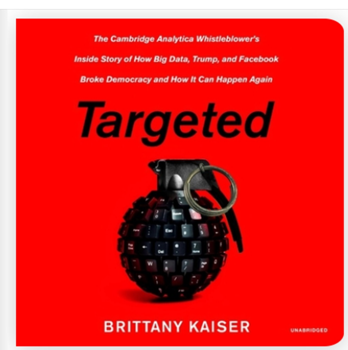
Why should Chemist develop some Math and Programming skills?

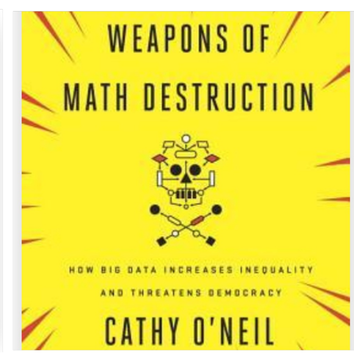












Students in the era of AI-Agents

SEPTEMBER 2 · 24 MINS

S3 EP4 - 5 tips for CAE engineers in the era of AI

The Neil Ashton Podcast

▶ Play

AUGUST 25 · 27 MINS LEFT

AI took your job

Today, Explained

▶ Resume

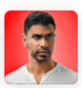
AUGUST 18 · 38 MINS LEFT

ChatGPT Brain Rot Debate: The Fastest Way to Get Dementia, Watch This Before Using ChatGPT Again, Especially If Your Kids Use It!

The Diary Of A CEO with Steven Bartlett

August 10

Jacob Tsimerman: He Won Math's Highest Prize. Then Announced the End



Theories of Everything with Curt Jaimungal >

August 29

S4 EP7 - Should You Still Study Engineering in the Age of AI?

The Neil Ashton Podcast >

Discussion on LLMs or AI-Agents

To what extend have you used LLMs for?

How much your day depends on them?

Do you like them?